

Ph.D. Thesis Proposal

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1 Title

MATHEMATICAL MODELLING AND APPLICATION OF OPTIMAL CONTROL
STRATEGIES TO SELECTED EPIDEMICS

2 Problem Statement

Epidemics call for serious concern in our world today. As such, it is imperative to employ every tool available to check their spread. Mathematical models have become very useful in investigating the spread and control of diseases like HIV/AIDS, Cancer, Hepatitis, Ebola, etc.

In order to effectively control the spread of epidemics, the dynamics of such epidemics have to be understood and necessary parameters have to be manipulated to get desired results.

3 Review of Related Literature

The highest outbreak of the Ebola Virus Disease (EVD) occurred in West Africa in 2014. The epidemic took a centre stage in the turn of events and it became a subject of great contention [1, 4, 10, 13]. Thousands of lives were lost, businesses crumbled and diplomatic ties were strained as a result of the deadly epidemic. The unusual outbreak of the EVD in West Africa can be explained by high density of populations [10]. Efforts aimed at curbing the spread of the disease can be forecast and modelled to provide useful information for policy makers [13].

Systems of coupled differential equations are used very commonly in modelling biological systems. The evolution of cells and the spread of epidemics can be so represented [2]. Such systems are investigated critically to discover the dynamics of the cases under consideration. Several studies revealed that the earlier view that carcinogenic tumour growth is merely a function of nutrition is not totally true. It has been discovered that a whole lot of complexity is involved in the process [15].

The spread of HIV has decimated populations since it was discovered in the 1980s. Certain lifestyles of people support this trend to a very large extent. Government budgets which should

have been spent on social amenities have been diverted to take care of the issues of HIV/AIDS [3]. [9] examines the effects of anti-retroviral treatment (ART) on an in-host model of HIV infection; mathematical analysis and bifurcation were carried out on the model and the study revealed the cause of a sudden rebound in viral load when ART ceases.

Being a serious threat to humans, [5] sought to understand the dynamics of infectious diseases. The study designed new mathematical models for the spread of such diseases and the impacts of interventions. [7] sheds light on the SIR (Susceptible-Infectives-Removed) model as the basis for modelling epidemics. The **susceptible** are those who are vulnerable; the **infectives** are those who are infected and can infect others while the **removed** are those who were infected but have been cured thereby being immune to the epidemic.

Optimal control has proven to be a very useful tool in controlling the spread of all manners of diseases [6, 12, 14]. An optimal control problem was formulated in [6] to study the optimal strategies for the medical treatment and quarantine of influenza viruses and structural results for the optimal strategies were obtained. [12] presents the application of optimal control theory to an SI malaria model with non-linear incidence rate. The study shows that an optimal controlled treatment strategy would ensure significant reduction in malaria incidence if fully adhered to. In [14], the optimal control of recurrent diseases were characterized through the use of prevention and treatment and useful policy-related discoveries were made. Optimal control theory is necessary as it helps us to understand how to channel our models towards desired ends. As such, it will be easy to advise policy makers on the best decisions to take in either managing or eradicating epidemics.

4 Research Objectives

This thesis is centred on developing a controlled evolving system for the spread of selected epidemics and to examine how the evolution takes place. An optimal control problem will be formulated, analysed and solved to determining the best ways of controlling the epidemics of interest. Recommendations will be made to relevant authorities for consideration.

5 Research Methodology

Our research objectives will be accomplished by:

- developing relevant epidemic models,
- computing the disease-free and endemic equilibrium solutions,
- determining the conditions for the stability of the equilibria,
- determining the basic reproduction numbers for the models,
- performing simulations to determine the impact of different control strategies,
- determining appropriate cost functionals associated with the use of various controls,
- formulating optimal control problems,
- proving the existence and uniqueness of the optimal controls,

- establishing the necessary conditions for optimality using the Pontryagin’s maximum principle,
- characterizing the optimal controls,
- solving the optimal control problems numerically,
- interpreting the solutions as optimal strategies for controlling the diseases.

References

- [1] Althaus, C.L. (2014), *Estimating the Reproduction Number of Ebola Virus (EBOV) During the 2014 Outbreak in West Africa*, PLOS Currents Outbreaks, Sep. 2: Edition 1.
- [2] Bellomo, N., Li, N.K. and Maini, P.K. (2008), *On the Foundations of Cancer Modelling: Selected Topics, Speculations, and Perspectives*, Mathematical Models and Methods in Applied Sciences, World Scientific Publishing Company, Vol. 18, No. 4:593-646.
- [3] Bozkurt, F. and Peker, F. (2014), *Mathematical Modelling of HIV Epidemic and Stability Analysis*, Advances in Difference Equations, 2014:95.
- [4] Chowell, G., Hengartner, N.W., Castillo-Chavez, C., Fenimore, P.W. and Hymann, J.M. (2004), *The Basic Reproductive Number of Ebola and the Effects of Public Health Measures: The Cases of Congo and Uganda*, Journal of Theoretical Biology, 229:119-126.
- [5] Chowell-Puente, G. (2005), *Mathematical Models of Emergent and Re-Emergent Infectious Diseases: Assessing the Effects of Public Health Interventions on Disease Spread*, A PhD Dissertation presented to the Faculty of the Graduate School of Cornell University, Ithaca, NY.
- [6] Elena, G. and Zhu, Q (2013), *Optimal Control of Influenza Epidemic Model with Virus Mutations*, 2013 European Control Conference (ECC), July 17-19, 2013, Zürich, Switzerland..
- [7] Iannelli, M. (2005), *The Mathematical Modeling of Epidemics*, Summer School on Mathematical Models in Life Science: Theory and Simulation, Dobbiaco, Bolzano-Italy.
- [8] Johnson, D., McKeever, S., Stamatakis, G., Dionysiou, D., Graf, N., Sakkalis, V., Marias, K., Wang, Z. and Deisboeck, T.S.(2013), *Dealing with Diversity in Computational Cancer Modeling*, Cancer Informatics, 12:115-124.
- [9] Li, M.Y. and Wang, L. (2014), *Backward Bifurcation in a Mathematical Model for HIV Infection in Vivo with Anti-Retroviral Treatment*, Nonlinear Analysis: Real World Applications, 17:147-160.
- [10] Majumder, M.S. (2015), *Mathematical Modeling of the 2014 Ebola Outbreak*, Health Map, Massachusetts Institute of Technology, Cambridge, MA.
- [11] Namawejje, H., Lubobi, L.S., Kuznetsov, D. and Wobudeya, E. (2014), *Modeling Optimal Control of Rotavirus Disease with Different Control Strategies*, J. Math. Comput. Sci. 4, No. 5:892-914.
- [12] Okosun, K.O. and Makinde, O.D. (2013), *Optimal Control Analysis of Malaria in the Presence of Non-Linear Incidence Rate*, Appl. Comput. Math., V.12, N.1, pp.20-32.

- [13] Rivers, C.M., Lofgren, E.T., Marathe, M., Eubank, S. and Lewis, B.L. (2015), *Modeling the Impact of Interventions on an Epidemic of Ebola in Sierra Leone and Liberia*, Network Dynamics and Simulation Science Lab, Virginia Bioinformatics Institute, Virginia Polytechnic Institute and State University, Blacksburg, VA.
- [14] Rowthornyaand, R. and Toxvaerdz, F. (2013), *The Optimal Control of Infectious Diseases via Prevention and Treatment*, Jel Classification, 2:73-118.
- [15] Roose, T., Chapman, S.J. and Maini, P.K. (2007), *Mathematical Models of Avascular Tumor Growth*, Society for Industrial and Applied Mathematics (SIAM) Review, Vol. 49, No. 2, pp. 179-208.